

CE Radio Test Report

Applicant : Magne AI Global tech limited
Equipment : MAG1
Brand Name : MaQ
Model Name : MA1
Standard : ETSI EN 303 417 V1.1.1 (2017-09)
Test Date(s) : Mar. 14, 2026 ~ May 21, 2026

The measurement shown in this test report is tested in accordance with the test procedures given in ETSI EN 303 417 V1.1.1 (2017-09).

The test results in this report apply exclusively to the tested model / sample. Without written approval of Sporton International Inc. (KunShan), the test report shall not be reproduced except in full.

Jason Jia

Approved by: Jason Jia



Sporton International Inc. (Kunshan)
No. 1098, Pengxi North Road, Kunshan Economic Development Zone
Jiangsu Province 215300, People's Republic of China



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History of this test report

Report No.	Version	Description	Issued Date
ER612311E	01	Initial issue of report	May 25, 2026

Summary of Test Result

Report Clause	Ref. Std. Clause	Test Items	Measured	Result (PASS/FAIL)
3.1	4.3.4	H-Field Strength	-3.82 dBuA/m@10m	Pass
3.2	4.3.2	Permitted range of operating frequencies	Permitted Range: F _L : 100 kHz F _H : 300 kHz	Pass
	4.3.3	Operating frequency ranges	OBW = 0.760 kHz	Pass
3.3	4.3.5	Transmitter spurious emissions	Under limit 18.89 dBm @ 0.1409 MHz	Pass
	4.3.6	Transmitter out of band (OOB) emissions	See report section 3.3.5	Pass
-	4.3.7	WPT system unwanted conducted emissions	N/A for cable length less than 3 m	N/A
3.4	4.4.2	Receiver blocking	See report section 3.4.5	Pass

Conformity Assessment Condition:

1. The test results (PASS/FAIL) with all measurement uncertainty excluded are presented against the regulation limits or in accordance with the requirements stipulated by the applicant/manufacture who shall bear all the risks of non-compliance that may potentially occur if measurement uncertainty is taken into account.
2. The measurement uncertainty please refer to each test result in the section "Measurement Uncertainty"

Disclaimer:

The product specifications of the EUT presented in the test report that may affect the test assessments are declared by the manufacturer who shall take full responsibility for the authenticity.



1 General Description

1.1 Applicant

Magne AI Global tech limited

FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.2 Manufacturer

Magne AI Global tech limited

FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.3 Product Feature of Equipment Under Test

Product Feature & Specification	
Product Name	MAG1
Brand Name	MaQ
Model Name	MA1
IMEI / SN Code	Conducted: MGUGLM54300121 Receiver spurious emissions: 016813000001816 for above 30MHz 016813000001816 for above 30MHz Blocking: MGUGLM54300121

1.4 Modification of EUT

No modifications are made to the EUT during all test items.

1.5 Information

1.5.1 RF General Information

RF General Information			
Frequency Range	Mode	Frequency (kHz)	H-Field Strength (dBuA/m)
100 -205 kHz	WPC	141.5	-3.82
Note 1: Field strength performed peak level at 10m. Note 2: For Inductive applications (e.g., animal identification, alarm systems, cable detection, waste management, personal identification, wireless voice links, access control, proximity sensors, data transfer to handheld devices, automatic article identification, wireless control systems, automatic road tolling and anti-theft systems including RF anti-theft induction systems.) Note 3: The device supports WPT RX only.			

1.5.2 Antenna Information

Antenna General Information		
No.	Ant. Cat.	Ant. Type
1	Integral	FPC

1.5.3 Type of EUT

Type of EUT	
☒	Stand-alone
☐	Combined (EUT where the radio part is fully integrated within another device) Combined Equipment - Brand Name / Model No.:
☐	Plug-in radio (EUT intended for a variety of host systems) Host System - Brand Name / Model No.:
☐	Other: The EUT place with the platform.

1.5.4 Test Signal Duty Cycle

Operated Mode for Worst Duty Cycle	
☒	Operated normally mode for worst duty cycle
☐	Operated test mode for worst duty cycle
Test Signal Duty Cycle (x)	
☒	100%

1.5.5 Test Mode

Spurious Emission	
Mode	Condition
1	Operating Mode + EUT with Wireless Charge Cradle + USB Cable (Charging from Adapter)
Conducted	
Mode	Condition
1	Operating Mode + EUT with Wireless Charge Cradle + USB Cable (Charging from Adapter)
Blocking	
Mode	Condition
1	Operating Mode + EUT with Wireless Charge Cradle + USB Cable (Charging from Adapter)

1.5.6 EUT Operational Condition

Supply Voltage	■	AC mains	□	DC		
Type of DC Source	■	External AC adapter	□	Internal DC supply	□	From System
Test Voltage	■	Vnom (3.87 V)	■	Vmax (4.45V)	■	Vmin (3.5 V)
Test Climatic	■	Tnom (20.0 °C)	■	Tmax (55.0 °C)	■	Tmin (-10 °C)

1.5.7 Overview of operational modes within a WPT system

Mode	
□	1. Base station in stand-by, idle mode 2.3.4
■	2. Communication before charging, adjustment charging mode / position
■	3. Communication
■	4. Energy transmission

Note: If the distance D is < 1 m, the WPT system shall be tested with the set-up described as mode 3.

1.5.8 Testing Applied Standards

According to the specifications of the manufacturer, the EUT must comply with the requirements of the following standards:

- ♦ ETSI EN 300 330 V2.1.1 (2017-02)
- ♦ ETSI EN 303 417 V1.1.1 (2017-09)

Note: All test items were verified and recorded according to the standards and without any deviation during the test.

**1.5.9 Testing Location Information**

Sporton International Inc. (Kunshan) is accredited to ISO/IEC 17025:2017 by American Association for Laboratory Accreditation with Certificate Number 5145.02.

Sporton International Inc. (KunShan)				
Address: No. 1098, Pengxi North Road, Kunshan Economic Development Zone, Jiangsu Province 215300 People's Republic of China				
Tel: +86-512-57900158				
Test Condition	Test Site No.	Test Engineer	Test Environment	Test Date
RF Conducted	TH01-KS	Smile	Temp: 21~25℃ Humidity: 51~54%	Mar. 19, 2026
Radiated Emission (9kHz~30MHz)	03CH02-KS	Fang	Temp: 21~23℃ Humidity: 41~42%	May 21, 2026
Radiated Emission (30MHz~1GHz)	05CH02-KS	Warm Wang	Temp: 21~23℃ Humidity: 41~43%	Mar. 14, 2026
Receiver Blocking	03CH02-KS	Fang	Temp: 21~23℃ Humidity: 41~42%	May 21, 2026

1.5.10 Measurement Uncertainty

Emissions test results be included in the report. The measurement uncertainties given below are based on a 95% confidence level (based on a coverage factor (k=2))

Measurement Uncertainty		
Test Item		Uncertainty
Radio Frequency		± 0.38ppm
RF power conducted		± 0.90 dB
Unwanted Emissions (for 03CH02-KS)	9 kHz – 30 MHz	± 3.30 dB
Unwanted Emissions (for 05CH02-KS)	30 MHz– 1000 MHz	± 2.54 dB
Temperature		±0.86 °C
Humidity		±0.022 %

2 Test Configuration of Equipment under Test

2.1 The Worst Case Modulation Configuration

Worst Modulation Used for Conformance Testing	
Mode	Field Strength (dBuA/m@10m)
WPC	-3.82

2.2 Test Channel Frequencies Configuration

Test Channel Frequencies Configuration	
Mode	Test Channel Frequencies (kHz)
WPC	141.5

2.3 The Worst Case Measurement Configuration

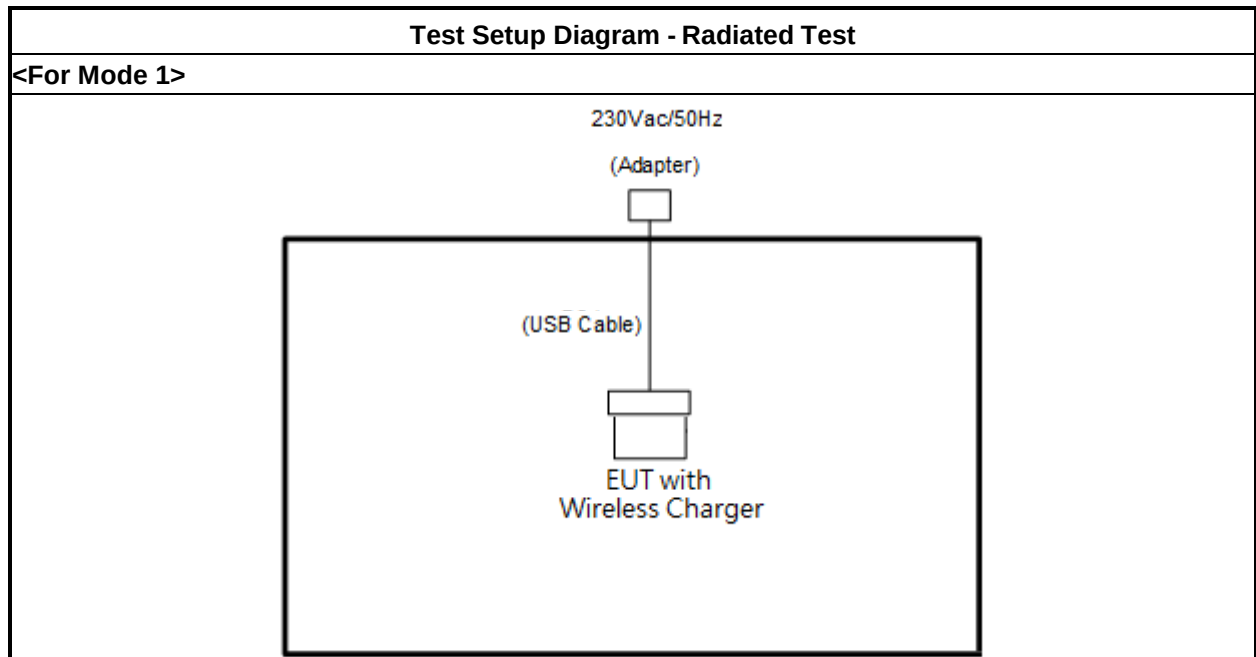
The Worst Case Mode for Following Conformance Tests	
Tests Item	H-Field Strength Permitted Range of Operating Frequencies Operating frequency range(s) (OFR)
Test Condition	Radiated measurement Test fixture for extreme conditions
Operating Mode	☒ 1. Adapter ☐ 2. Other:

Note: H-Field Strength is based on the worst adapter result of 10m chamber.

The Worst Case Mode for Following Conformance Tests			
Tests Item	WPT system unwanted radiated emissions		
Test Condition	Radiated measurement Receivers used in combination with permanently co-located transmitters continuously transmitting. Co-located is defined as < 3 m, then receiver spurious radiation do not apply to receivers used in combination with permanently co-located transmitters continuously transmitting.		
Operating Mode	Operating ☒ 1. Adapter mode (Y Plane)		
Orthogonal Planes of EUT	X Plane	Y Plane	Z Plane
			

The Worst Case Mode for Following Conformance Tests	
Test Items	Receiver Blocking
Test Condition	Conducted measurement at a receiver chain

2.4 The Worst Test Setup Diagram



2.5 EUT Operation Test Setup

WPT Charging.

3 Transmitter Test Result

3.1 H-Field Strength

3.1.1 H-Field Strength Limit

H-Field Strength Limit		
Frequency Range	H-Field Strength Limit [dBuA/m@10m]	Comments
19-21 kHz	72	
59-61 kHz	69.1 descending 10 dB/dec above 59 kHz	See note 1
79-90 kHz	67.8 descending 10 dB/dec above 79 kHz	See note 2
100-119 kHz	42	
119-135 kHz	66 descending 10 dB/dec above 119 kHz	See note 1
135-140 kHz	42	
140-148.5 kHz	37.7	
148.5-300 kHz	-5	
6.765-6.795 MHz	42	

NOTE 1: Limit is 42 dBuA/m for the following spot frequencies: 60 kHz $\pm$ 250 Hz and 129,1 kHz $\pm$ 500 Hz.
 NOTE 2: At the time of preparation of the present document the feasibility of increased limits for high power wireless power transmission systems to charge vehicles [i.4] was prepared. New specific requirements for such systems (e.g. higher H-field emission limits in the 79 - 90 kHz band) will be reflected within a future revision of the present document.

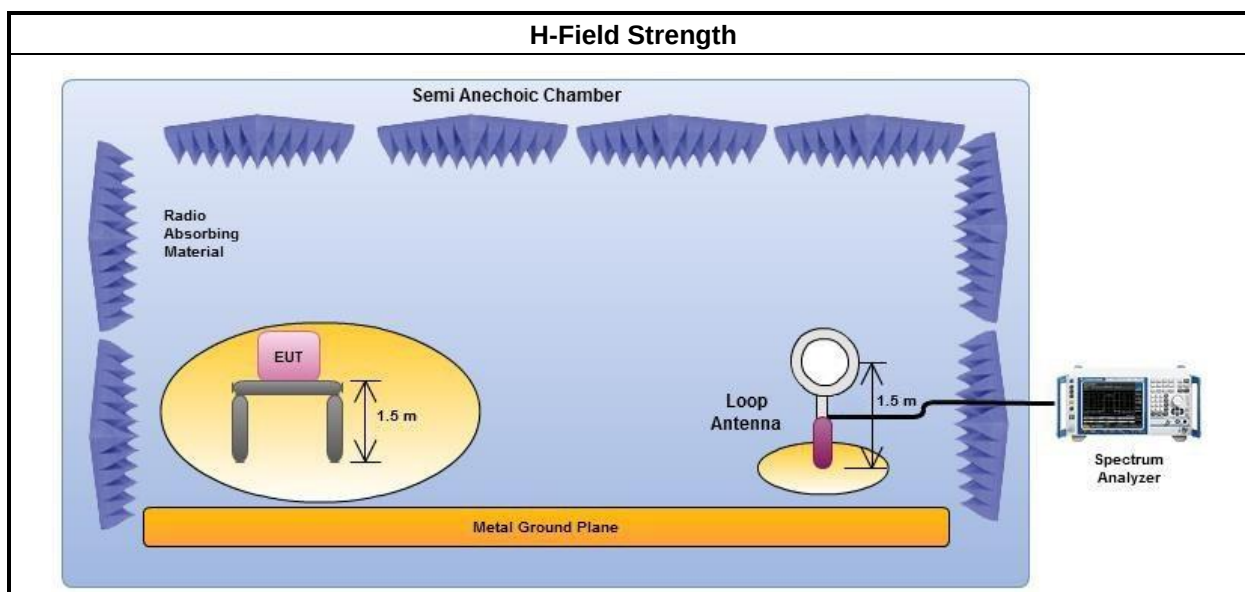
3.1.2 Measuring Instruments

Refer a test equipment and calibration data table in this test report.

3.1.3 Test Procedures

Test Method	
■ The measurements shall be performed at both normal environmental conditions and at the extremes of the operating temperature range. ■ Refer as EN 300 330, clause 6.2.4 for H-field measurement.	□ EUT had been measured at standard distance [10 m].
	■ EUT had been measured at another distance [3 m]. According to EN 300 330 Annex H2: $H_{3m} = H_{10m} + C_3$
	■ For extremes conditions, the EUT with test fixture shall then be related to the field strength measured in radiated condition. The compensation calculation will be used to calculate the absolute level of the extremes conditions.
	■ Measurement worst emissions of receive antenna polarization: Horizontal

3.1.4 Test Setup



3.1.5 Test Results

Transmitter H-Field Output Levels Result						
Condition	Mode	Freq. (kHz)	E-field (dBuV/m at 3m)	E-field (dBuV/m at 10m)	H-field (dBuA/m at 10m)	H-field Limit (dBuA/m at 10m)
TnomVnom	WPC	141.5	78.91	47.61	-3.89	37.7
TminVmax	WPC	141.5	78.95	47.65	-3.85	37.7
TminVmin	WPC	141.5	78.98	47.68	-3.82	37.7
TmaxVmax	WPC	141.5	78.79	47.49	-4.01	37.7
TmaxVmin	WPC	141.5	78.83	47.53	-3.97	37.7
Test Result: Complied						
Remark:						
1. Test is performed at 3m distance then convert to 10m results according to EN 300 330-1 Annex H.2.						
2. H-field(dBuA/m)=E-field limit(dBuV/m) - 51.5dB.						

3.2 Permitted range of operating frequencies and Operating frequency ranges

3.2.1 Permitted Range of Operating Frequencies Limit

Permitted Range of Operating Frequencies & OFR Limit	
The permitted range of operating frequency range(s) for intentional emissions shall be within 19 - 21 kHz, 59	
- 61 kHz, 79 - 90 kHz, 100 - 300 kHz, 6 765 - 6 795 kHz, see EN 303 417 Table 1.	

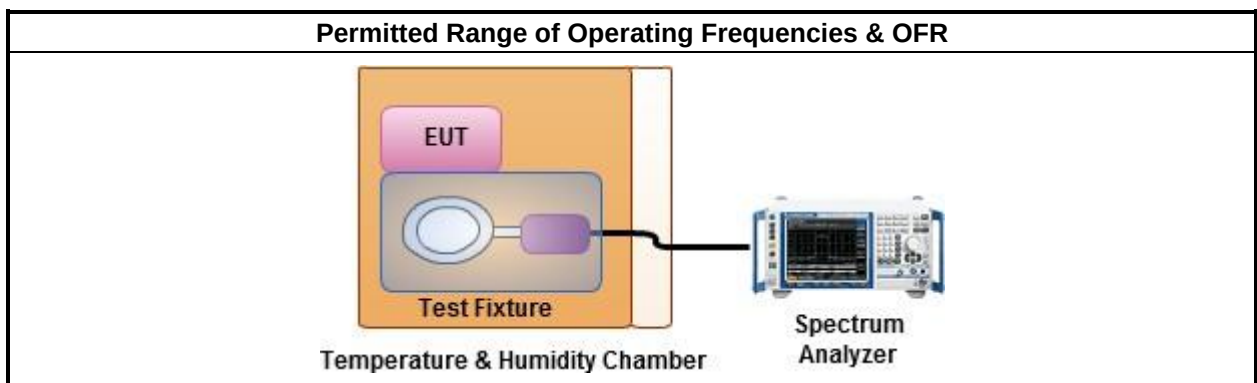
3.2.2 Measuring Instruments

Refer a test equipment and calibration data table in this test report.

3.2.3 Test Procedures

Test Method	
■	Refer as EN 300 330, clause 6.2.2 for permitted range of operating frequencies
■	Operating frequencies (FL and FH) below spurious limit and operating frequencies (FL and FH) fall in authorized band.
■	OFR shall be calculated as $BW = FH - FL$, and the centre frequency as : $FC = (FH + FL)/2$
■	Measurement worst emissions of receive antenna polarization: H (Horizontal).

3.2.4 Test Setup



3.2.5 Test Result of Permitted Range of Operating Frequencies

Permitted Range of Operating Frequencies Result							
Condition	Modulation Mode	Freq. (kHz)	Threshold Level	F _L (kHz)	F _H (kHz)	F _L Limit (kHz)	F _H Limit (kHz)
TnomVnom	WPC	141.56	OBW	141.255	142.015	100	300
TminVmax	WPC	141.56	OBW	141.255	142.015	100	300
TminVmin	WPC	141.71	OBW	141.255	142.015	100	300
TmaxVmax	WPC	141.56	OBW	141.255	142.015	100	300
TmaxVmin	WPC	141.71	OBW	141.255	142.015	100	300

Note: The frequency of charging will shift a little.

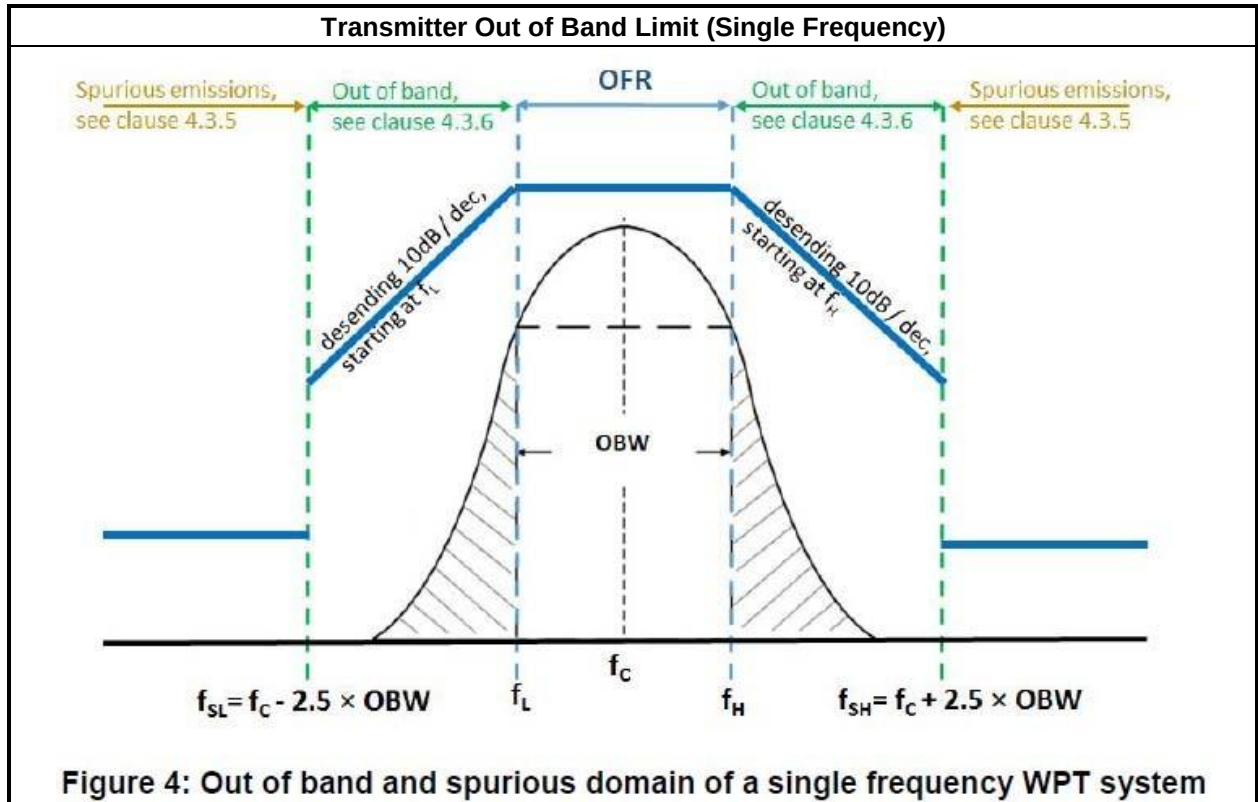
3.2.6 Test Result of Operating frequency ranges

Operating Frequency Range(s) (OFR) Result			
Condition	Modulation Mode	Freq. (kHz)	99% Modulation Bandwidth (kHz)
TnomVnom	WPC	141.56	0.760
TminVmax	WPC	141.56	0.760
TminVmin	WPC	141.71	0.760
TmaxVmax	WPC	141.56	0.760
TmaxVmin	WPC	141.71	0.760
Result		Complied	

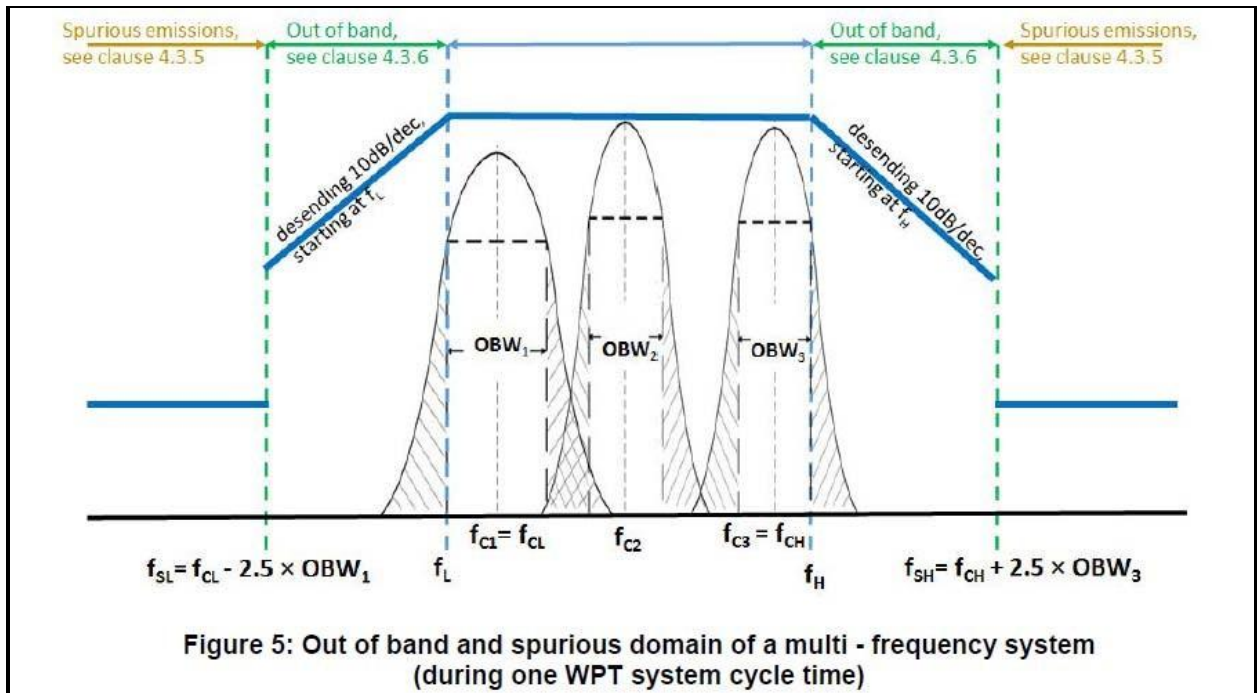
Note: The frequency of charging will shift a little.

3.3 Transmitter Spurious and Transmitter Out of Band (OOB) emissions

3.3.1 Transmitter Out of Band (OOB) Emissions Limit



Transmitter Out of Band Limit (Multi Frequency)



Transmitter Spurious and Out of Band Emissions (9 kHz - 30 MHz) H-field dBuA/m at 10 m		
Frequency State	Frequency 9 kHz ≤ f < 10 MHz	Frequency 10 MHz ≤ f < 30 MHz
Operating	27 dBuA/m at 9 kHz descending 10 dB/dec	-3.5 dBuA/m
Standby	5.5 dBuA/m at 9 kHz descending 10 dB/dec	-25 dBuA/m

Transmitter Spurious and Out of Band Emissions (30 MHz - 1 GHz)		
Frequency State	47 MHz to 74 MHz 87,5 MHz to 118 MHz 174 MHz to 230 MHz 470 MHz to 790 MHz	Other Frequencies between 30 MHz to 1 000 MHz
Operating	4 nW (-54 dBm)	250 nW (-36 dBm)
Standby	2 nW (-57 dBm)	2 nW (-57 dBm)

NOTE: "Operating" means mode 2, 3 and 4 according to Table 2; "standby" means mode 1 according to EN 303 417 Table 2.

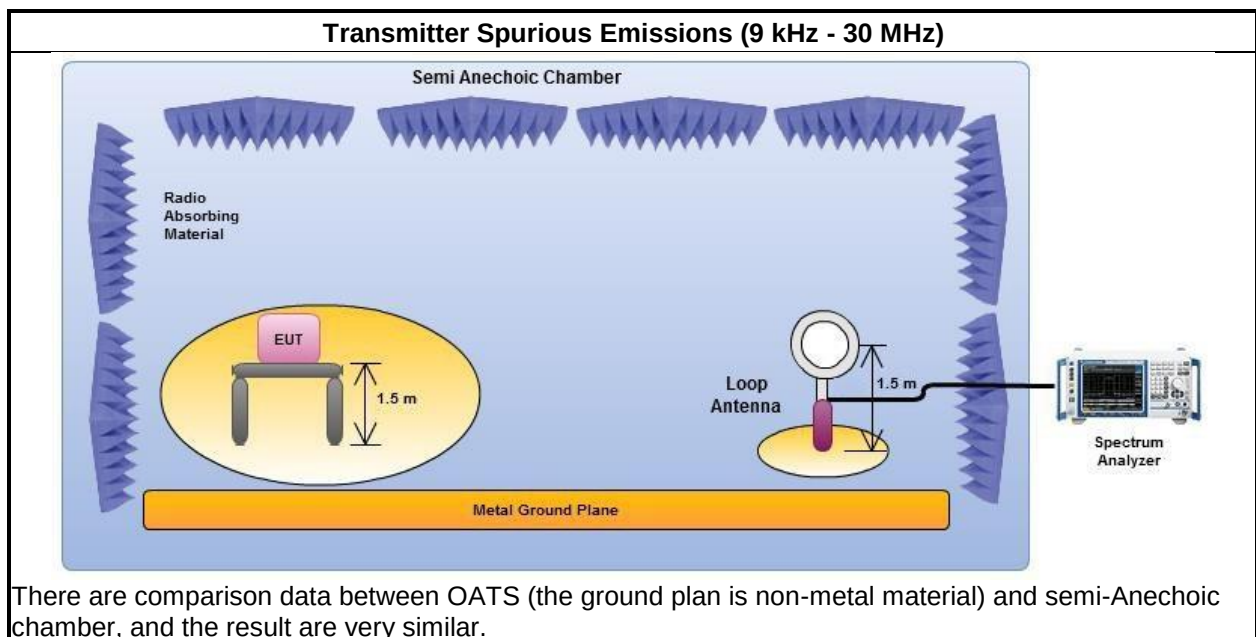
3.3.2 Measuring Instruments

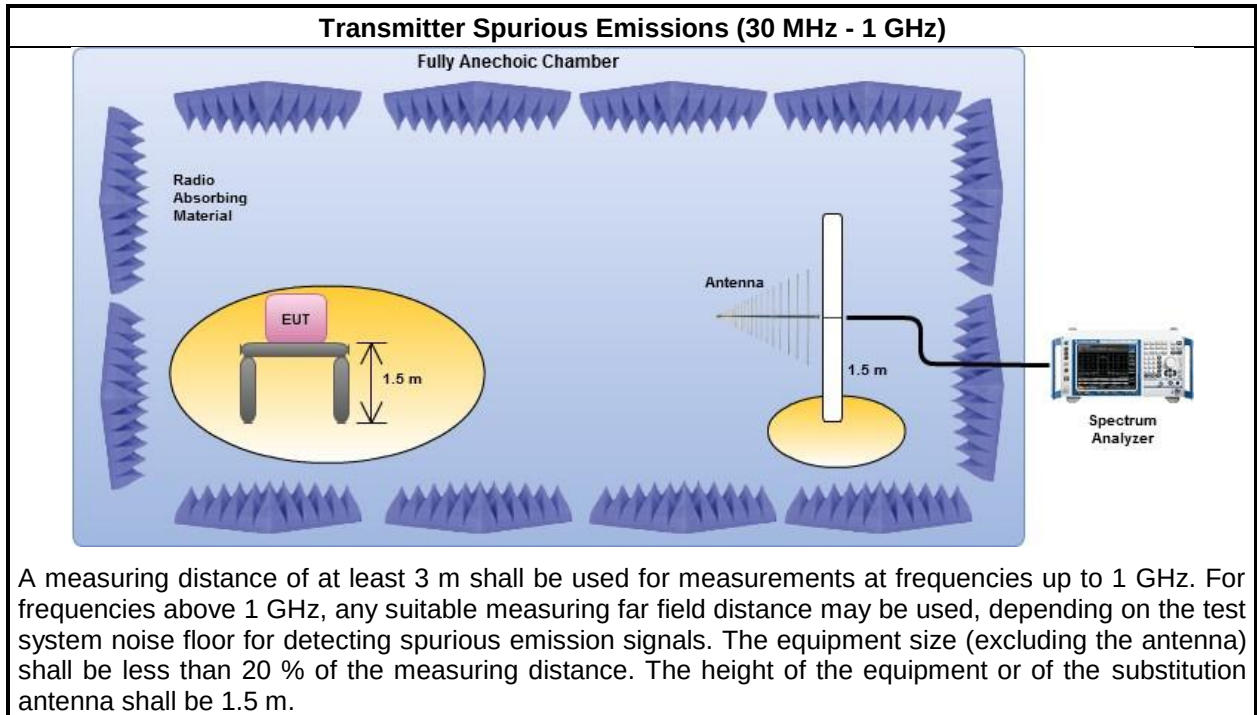
Refer a test equipment and calibration data table in this test report.

3.3.3 Test Procedures

Test Method	
■	Refer as EN 300 330, clause 6.2.7 for conducted measurement
■	Out of band emissions for single frequency.
□	Out of band emissions for multi frequency.
■	Refer as EN 300 330, clause 6.2.8, 6.2.9 for radiated measurement
□	If an unmodulated carrier cannot be obtained.
■	If EUT modulated by the normal test signal.
■	If standby mode is available.
□	If receivers used in combination with permanently co-located transmitters continuously transmitting. Co-located is defined as < 3 m, then receiver spurious radiation do not apply to receivers used in combination with permanently co-located transmitters continuously transmitting.
■	Measurement worst emissions of receive antenna polarization: H (Horizontal).

3.3.4 Test Setup





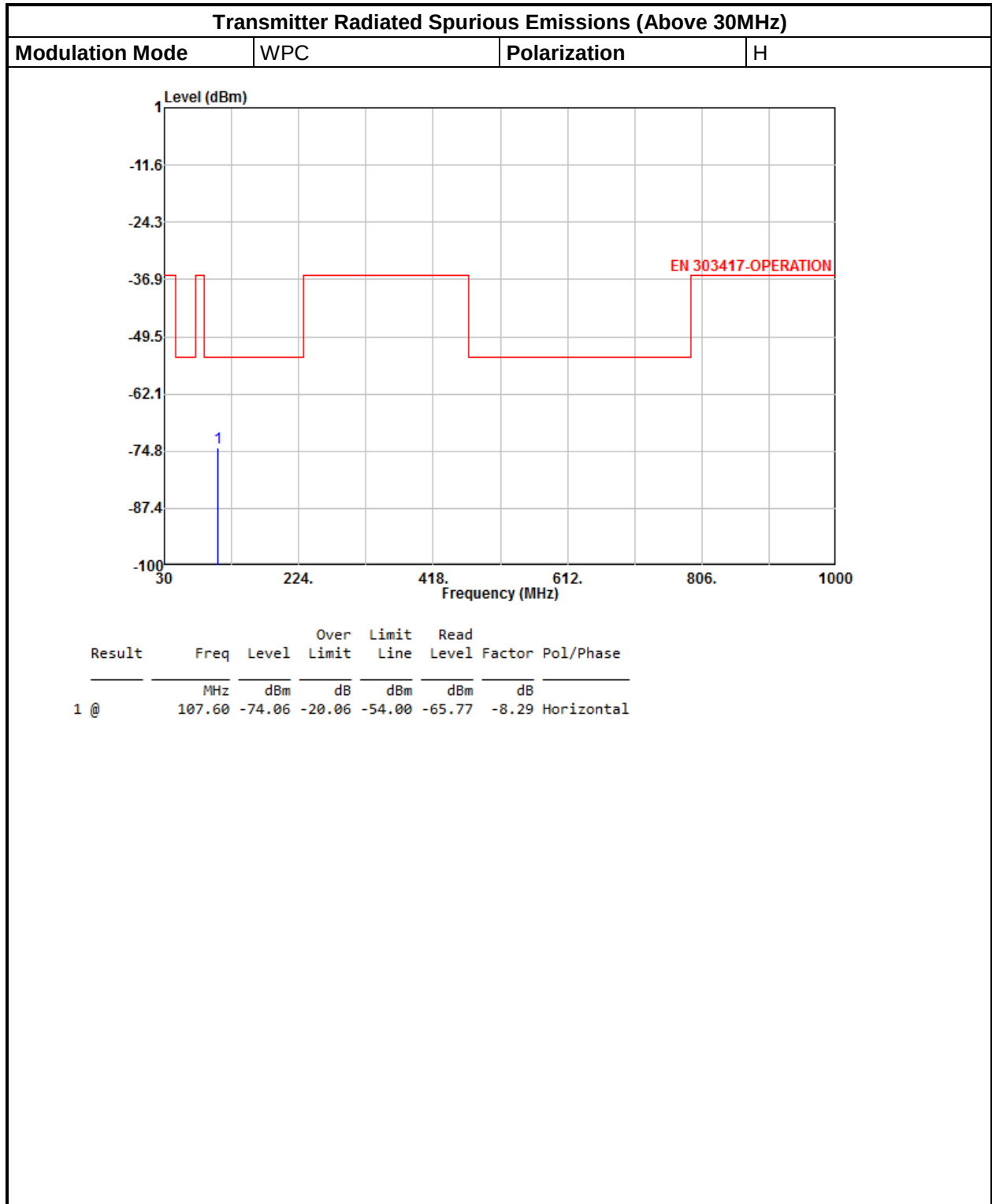
3.3.5 Transmitter Out of Band Emissions

Transmitter Out of Band Emission							
Mode	OOB						
Emission Frequency (MHz)	Polarization	Level (dBuV/m @10m)	E-Field (dBuV/m @10m)	E-Field Limit (dBuV/m @10m)	Level Type	Margin (dB)	Result
0.1409	H	78.96	47.66	66.5500	Peak	-18.89	Pass
0.1415	H	78.91	47.61	66.5300	Peak	-18.92	Pass
0.1421	H	78.65	47.35	66.5200	Peak	-19.17	Pass
0.1409	V	71.58	40.28	66.5500	Peak	-26.27	Pass
0.1415	V	71.54	40.24	66.5300	Peak	-26.29	Pass
0.1421	V	71.28	39.98	66.5200	Peak	-26.54	Pass
Remark: 1. Test is performed at 3m distance then convert to 10m results according to EN 300 330-1 Annex H.2. 2. E-field limit(dBuV/m)=H field limit(dBuA/m) + 51.5dB							

3.3.6 Transmitter Radiated Unwanted Emissions (Below 30MHz)

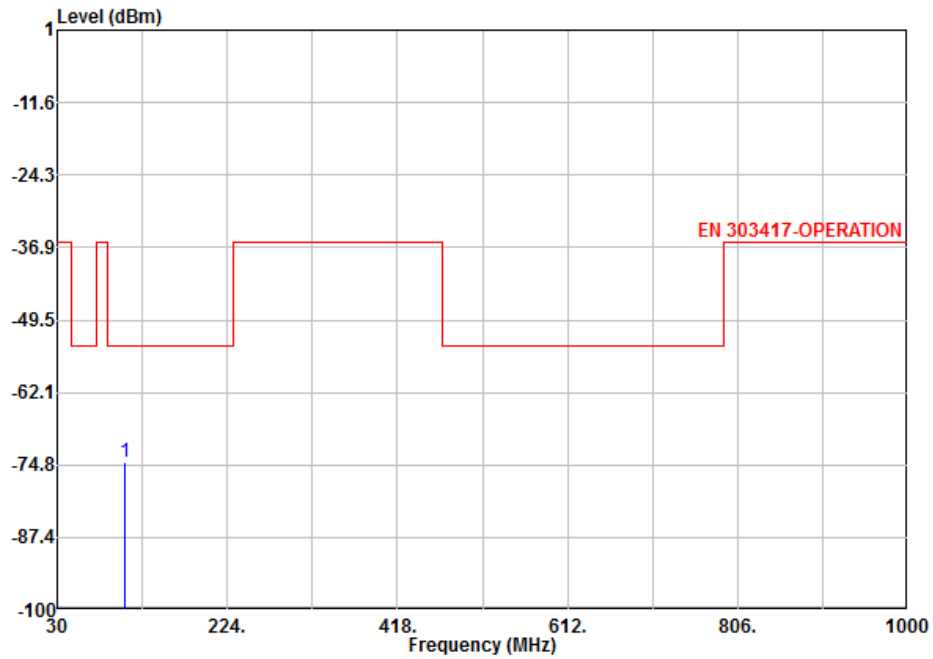
Transmitter Out of Band Emission							
Mode	TX						
Emission Frequency (MHz)	Polarization	Level (dBuV/m @10m)	E-Field (dBuV/m @10m)	E-Field Limit (dBuV/m @10m)	Level Type	Margin (dB)	Result
0.0630	H	51.81	20.51	71.00	Peak	-50.49	Pass
0.2396	H	57.19	25.89	61.82	Peak	-35.93	Pass
0.0500	V	46.90	15.60	70.99	Peak	-55.39	Pass
0.5381	V	53.97	22.67	61.82	Peak	-39.15	Pass
Remark: 1. Test is performed at 3m distance then convert to 10m results according to EN 300 330-1 Annex H.2. 2. E-field limit(dBuV/m)=H field limit(dBuA/m) + 51.5dB							

3.3.7 Transmitter Radiated Unwanted Emissions (Above 30MHz)



Transmitter Radiated Spurious Emissions (Above 30MHz)

Modulation Mode	WPC	Polarization	V
-----------------	-----	--------------	---



Result	Freq	Level	Over	Limit	Read		
	MHz	dBm	Limit	Line	Level	Factor	Pol/Phase
			dB	dBm	dBm	dB	
1 @	106.63	-74.17	-20.17	-54.00	-66.62	-7.55	Vertical

3.4 Receiver Blocking

3.4.1 Receiver Blocking Limit

Receiver Blocking Limit			
	In-Band Signal	OOB Signal	Remote-Band Signal
Frequency	Centre Frequency(f_c) of the WPT system	$f = f_c \pm F$ (see note)	$f = f_c \pm 10 \times F$ (see note)
Signal Level Field Strength at the EUT	72dBuA/m	72dBuA/m	82dBuA/m
NOTE: $F = \text{OFR}$			

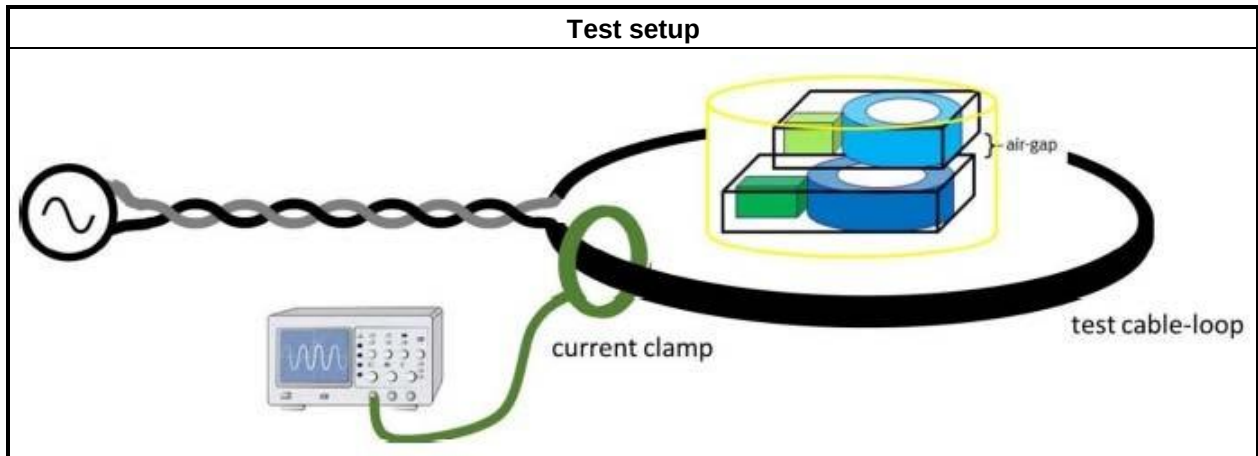
3.4.2 Measuring Instruments

Refer a test equipment and calibration data table in this test report.

3.4.3 Test Procedures

Test Method	
Refer as EN 303 417, clause 6.3.2 for receiver blocking. Test cable-loop $R=1$ m	
	In-band signal: f_c.
	OOB signal : $f=f_c \pm \text{OFR}$
	Remote-band signal : $f=f_c \pm 10 \times \text{OFR}$

3.4.4 Test Setup



**3.4.5 Test Result of Receiver Blocking**

Center Frequency (kHz)	Frequency (kHz)	I (dBuA)	R (m)	Blocking H field (dBuA/m)	Limit H field (dBuA/m)	Result
141	141.5	81.659	1	85.638	72	Complies
	140.89219	81.641	1	85.620	72	Complies
	142.10781	81.669	1	85.648	72	Complies
	135.4219	81.438	1	85.417	82	Complies
	147.5781	81.722	1	85.701	82	Complies

OFR(kHz): 0.760

4 Test Setup Photographs

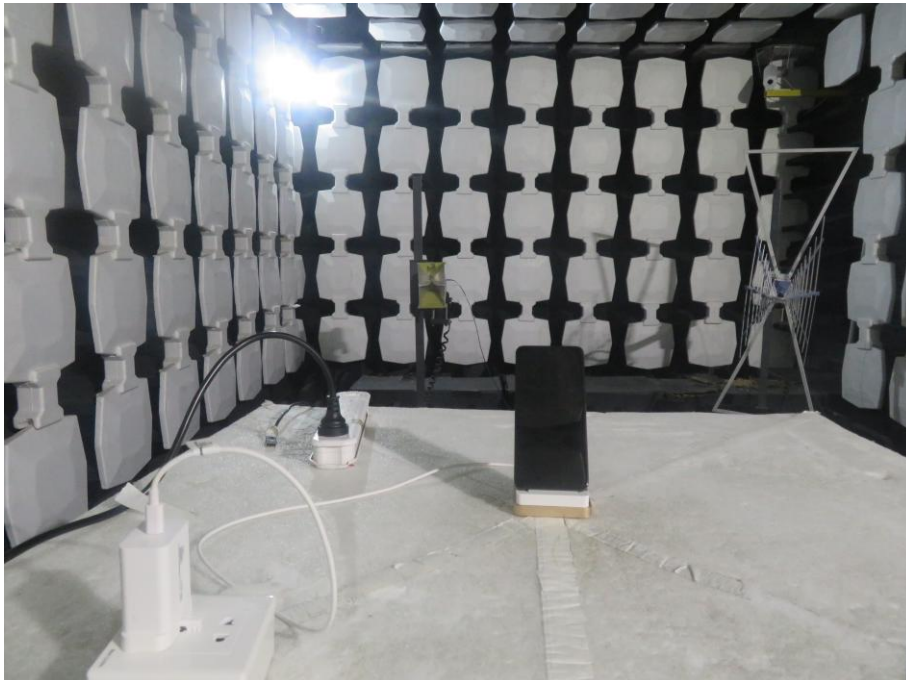
<RSE 9kHz~30MHz>

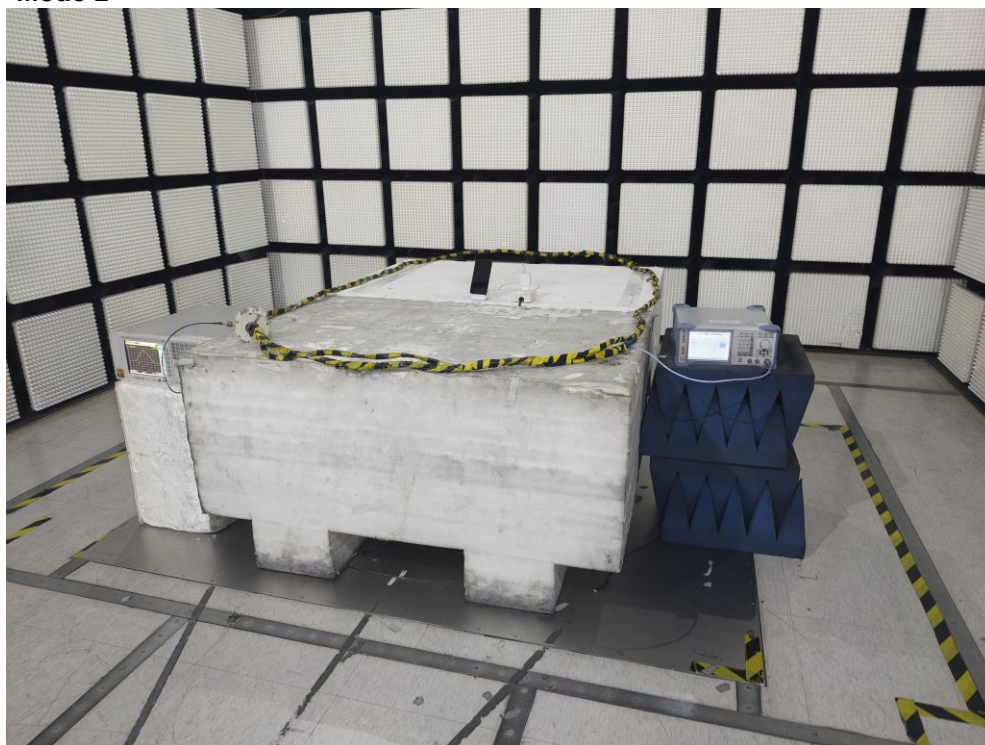
Mode1



<RSE 30MHz~1GHz>

Mode 1



<Receiver Blocking>**Mode 1**



5 Test Equipment and Calibration Data

Instrument for Conducted Test

Instrument	Manufacturer	Model No.	Serial No.	Characteristics	Calibration Date	Calibration Due Date
Spectrum Analyzer	R&S	FSV40	101040	10Hz~40GHz	Oct. 09, 2025	Oct. 08, 2026
Temperature & humidity chamber	Hongzhan	LP-150U	H2014011440	-40~+150°C 20%~95%RH	Jul. 03, 2025	Jul. 02, 2026
DC Power Supply	GW INSTEK	GPD-2303S	GEO861339	Max 31	Jan. 01, 2025	Dec. 31, 2026

Instrument for Radiated Emissions (9kHz~30MHz)

Instrument	Manufacturer	Model No.	Serial No.	Characteristics	Calibration Date	Calibration Due Date
EMI Test Receiver	R&S	ESR7	101403	9kHz~7GHz;Max 30dBm	Oct. 09, 2025	Oct. 08, 2026
Loop Antenna	R&S	HFH2-Z2	100321	9kHz~30MHz	Sep. 08, 2025	Sep. 07, 2026

Instrument for Radiated Emissions (30MHz~1GHz)

Instrument	Manufacturer	Model No.	Serial No.	Characteristics	Calibration Date	Calibration Due Date
Signal and spectrum analyzer	Rohde & Schwarz	FSV3044	101177	10Hz~44GHz	Jul. 03, 2025	Jul. 02, 2026
Log-periodic antenna	R&S	HL562	100446	30MHz~3GHz	Jan. 21, 2026	Jan. 20, 2027
Amplifier	SONOMA	310N	372171	9KHz ~1GHZ	Jan. 02, 2026	Jan. 01, 2027

Instrument for Receiver Blocking

Instrument	Manufacturer	Model No.	Serial No.	Characteristics	Calibration Date	Calibration Due Date
EXA Spectrum Analyzer	KEYSIGHT	N9010B	MY59071191	10Hz~44GHz	Apr. 02, 2026	Apr. 01, 2027
Current probe	TESEQ	CIP 9136A	48129	10KHz~400MHz	Jun. 23, 2025	Jun. 22, 2026
Signal Generator	R&S	SMBV100A	258305	9KHz~6GHz	Jan. 06, 2026	Jan. 05, 2027
Blocking test cable loop	N/A	N/A	WPCLOOPCAB LE001	Radius=1m	NCR	NCR

———— THE END ————